MR. ALLEN'S MATH CLASS · IC IMAGINE

Design-a-Drink Company Project

Unit 9 · Volume of Cylinders, Cones & Spheres · NC.8.G.9

Name: ________________________________ **Class Period:** ________ **Due Date:** _____________

The Mission

You just founded a beverage company! Design three containers for your new drink — one **cylinder** (a can), one **cone** (a party cup), and one **sphere** (a fun promo container). Your job is to calculate exactly how much each one holds so the factory can fill them.

Requirements

- **Brand It (15 pts):** Invent a company name, a drink name, and a logo. Draw all three containers with your branding.
- **Dimension It (15 pts):** Choose realistic measurements for each container (radius and height, in centimeters) and label them clearly on your drawings.
- **Calculate It (45 pts):** Find the volume of each container. Show every step: write the formula, substitute your numbers, then solve. Use 3.14 for π and round to the nearest tenth. Formulas: Cylinder $V = \pi r^2 h$ · Cone $V = \frac{1}{3}\pi r^2 h$ · Sphere $V = \frac{4}{3}\pi r^3$
- **Compare It (15 pts):** Rank your three containers from largest to smallest volume. Then answer: if the cone and cylinder have the same radius and height, exactly how many cone-fulls fill the cylinder? Explain why.
- **Price It (10 pts):** Your drink costs 0.5¢ per cubic centimeter to make. Calculate the production cost of each container.

Show-Your-Work Table

Container	r (cm)	h (cm)	Formula with numbers substituted	Volume (cm^3)
Cylinder				
Cone				
Sphere				

Grading Rubric

Category	4 — Excellent	3 — Good	2 — Developing	1 — Beginning
Branding & drawings	All 3 drawn, labeled, creative	All 3 drawn and labeled	2 drawn or unlabeled	0–1 drawn

Cylinder volume	Correct with all steps	Correct, few steps	Minor error	Missing or incorrect
Cone & sphere volume	Both correct with steps	One correct with steps	Both attempted, errors	Missing
Compare & price	Ranking, cone answer, and costs all correct	One part has an error	Two parts have errors	Mostly missing

Bonus (+5): Design a fourth container that combines two shapes (like a cylinder with a cone top) and find its total volume.